ROBOTIC VEHICLE FOR SURVEILLANCE AND HAZARDOUS ZONE OPERATIONS

CONTENTS

1. Abstract
2. Objective
3. Introduction
4. Literature Review
5. Problem Identification
6. Existing System
7. Proposed System
8. Hardware Requirements
9. Software Requirements
10. Output
11. Advantages
12. Applications
13. Conclusion
14. References

ABSTRACT

- Border surveillance and hazardous zone monitoring require continuous observation while exposing personnel to severe risks.
- This project presents an embedded robotic vehicle designed for remote surveillance using IoT and mobile based visual monitoring.
- The robot employs DC gear motors controlled through an L293D motor driver for smooth multidirectional movement.
- An IoT enabled controller allows cloud based remote operation, real time sensor monitoring, and instant alerts.
- Fire detection and human presence identification enhance safety and security. A mobile camera provides live visual feedback.
- The system minimizes human involvement and improves security operations through reliable, scalable, and cost effective embedded robotic architecture.

OBJECTIVE

- The primary objective of this project is to design and implement an embedded robotic vehicle capable of performing border surveillance and hazardous zone monitoring with minimal human intervention.
- The system aims to enhance security by enabling remote navigation, real time environmental sensing, and continuous visual monitoring using IoT technology.
- It focuses on detecting fire hazards and unauthorized human presence in restricted areas while transmitting data instantly to a cloud platform.
- The objective further includes developing a cost effective, scalable, and reliable robotic solution that improves situational awareness, supports decision making, and reduces risks to security personnel operating in dangerous environments.

INTRODUCTION

- Rising security concerns and hazardous working conditions have increased the demand for automated surveillance systems.
- Conventional border monitoring methods rely heavily on human patrols, exposing personnel to threats such as intrusions, fire hazards, and hostile environments.
- Recent advancements in robotics, embedded systems, and IoT have enabled intelligent surveillance platforms capable of operating remotely.
- Robotic vehicles integrated with sensors and cameras can perform continuous monitoring without fatigue.
- IoT connectivity further allows real time data transmission and remote control. This project introduces an embedded robotic vehicle that combines mobility, sensing, and cloud based monitoring to enhance border security and hazardous zone surveillance while reducing human risk.

LITERATURE REVIEW

S.NO	Author(s)	Title & Name of the Journal	Year	Remarks
1	Akilan T, Satyam Chaudhary, Princi Kumari and Utkarsh Pandey.	Surveillance Robot in Hazardous Place Using IoT & IEEE Xplore.	2025	• IoT mobile robot for live streaming and hazardous area monitoring.
2	Sufyan Danish, Samee Ullah Khan, Muhammad Attique Khan, Minh Dang L and Yahya Zweir.	Vision-based Fire Management System Using UAVs & Springer Nature Link.	2025	• UAV-based fire detection using visual analytics and AI.
3	Thankappan M, Neetha K Nataraj and Alen Joby.	AI-Enabled Robotic Surveillance System & IJRTI.	2025	• Autonomous robotic system for critical infrastructure anomaly detection.

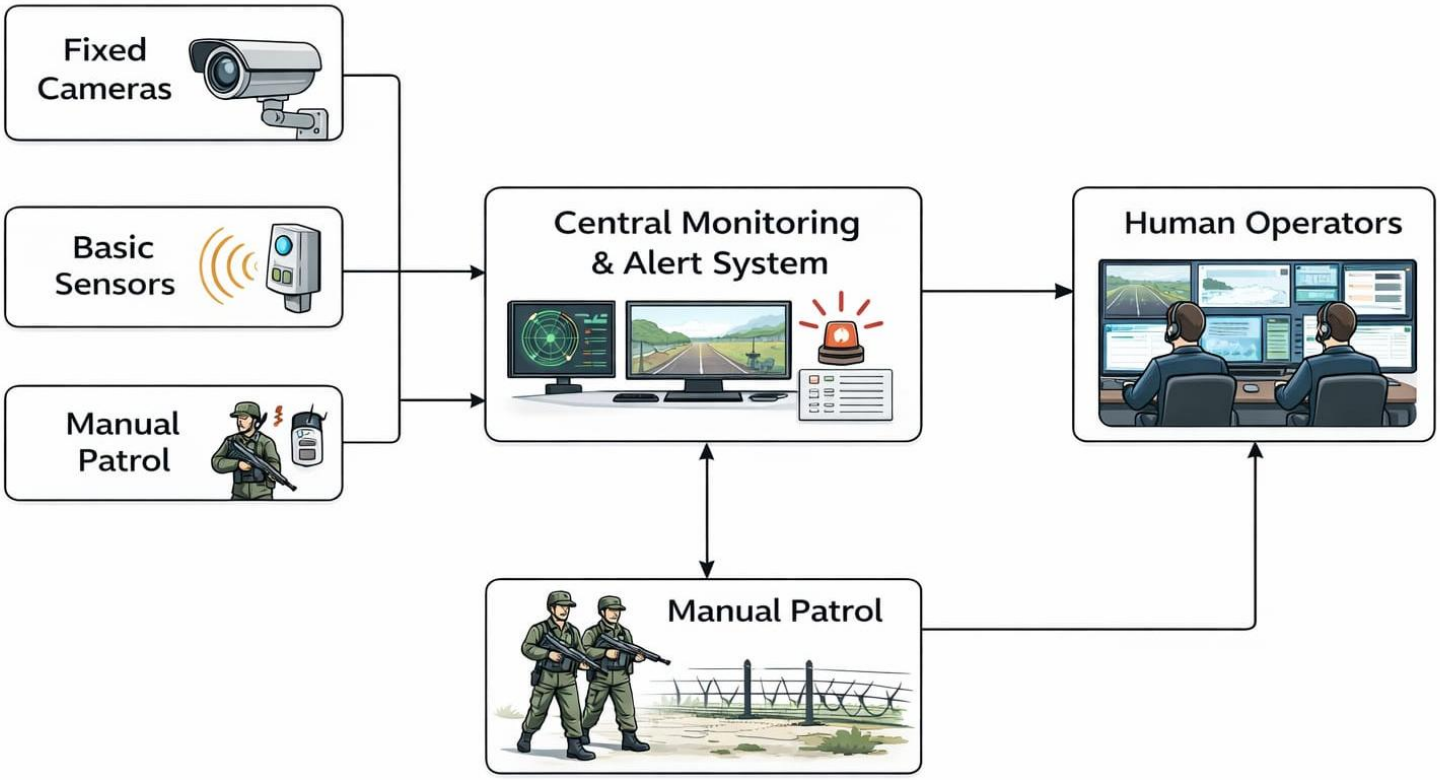
LITERATURE REVIEW

S.NO	Author(s)	Title & Name of the Journal	Year	Remarks
4	Mominur R Meem, Farah Alam Mim and Partho Sharothi Chowhan.	Artificially Intelligent Surveillance and Security Sentinel & IJEM.	2024	• Autonomous mobile robot with night vision and human detection.
5	Jyotsna S, Gawai and Prajakta Shende.	IoT-Based Wireless Surveillance Robot & IJRASET.	2024	• IoT robot integrating sensors for remote security monitoring.

PROBLEM IDENTIFICATION

- High Fatality Risk: Personnel are frequently exposed to life-threatening dangers like snipers, ambushes, and IEDs in active border zones.
- Lethal Environments: Hazardous areas involving chemical leaks, radiation, or structural instability (collapsed mines/buildings) are often "suicide missions" for humans.
- Surveillance Fatigue: Human attention spans drop significantly during long, monotonous shifts, leading to critical security lapses and missed breaches.
- Limited Sensory Perception: Human vision is restricted by total darkness, heavy fog, or thick smoke—conditions where intruders or hazards are most likely to exist.
- Communication Delays: Significant time-lag occurs between a human scout spotting a threat and the Command Center receiving an accurate report.

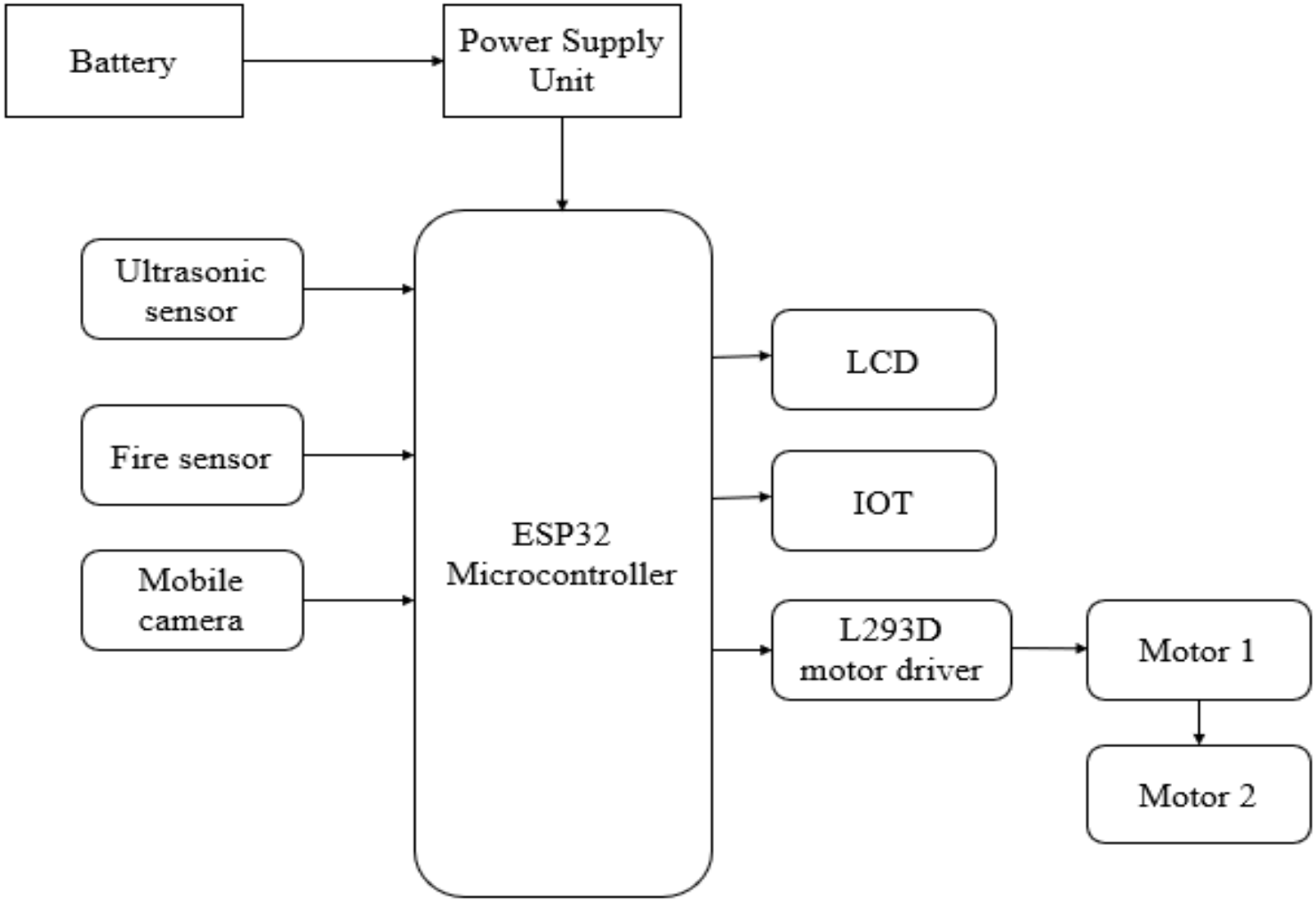
EXISTING SYSTEM



EXISTING SYSTEM WORKING

- Reliance on Static Infrastructure: Uses fixed cameras and basic sensors, creating potential blind spots and limiting the field of view to specific stationary locations.
- Manual Surveillance & Patrolling: Significant dependence on ground personnel for physical patrolling, which is labor-intensive and limits the frequency of area coverage.
- Centralized Monitoring Hub: All data streams (video, sensor alerts, and radio reports) converge at a Central Monitoring & Alert System for processing.
- Human-Centric Decision Making: Heavily relies on human operators to monitor screens in real-time and interpret data, increasing the risk of fatigue-related oversight.
- Reactive Response Framework: The system is designed to alert operators after a sensor is triggered or a visual is caught.

PROPOSED SYSTEM



PROPOSED SYSTEM WORKING

- The battery supplies power to the power supply unit. The power supply regulates the voltage and provides stable power to the ESP32 and other components.
- After powering ON, the ESP32 initializes all connected sensors and modules. It continuously monitors inputs from the ultrasonic sensor, fire sensor, and camera.
- The ultrasonic sensor measures the distance between the robot and nearby obstacles. If an object is detected within a set limit, the ESP32 decides the movement action.
- The fire sensor detects the presence of flame in the environment. When fire is detected, the ESP32 triggers alerts and sends notifications through IoT.

PROPOSED SYSTEM WORKING

- The mobile camera captures live images or video of the surroundings. This visual data can be monitored remotely through IoT connectivity.
- Based on sensor data, the ESP32 sends control signals to the L293D motor driver. The driver controls the speed and direction of the motors.
- Motor 1 and Motor 2 enable forward, backward, left, and right movement. This allows obstacle avoidance and smooth navigation of the robot.
- The LCD displays real-time sensor readings and system status. At the same time, the ESP32 transmits data to the cloud for remote monitoring.

HARDWARE REQUIREMENTS

- Battery
- Power Supply Unit
- ESP32 Microcontroller
- Ultrasonic Sensor
- Fire Sensor
- Mobile Camera
- LCD Display
- L293D Motor Driver
- DC Motor 1
- DC Motor 2

SOFTWARE REQUIREMENTS

- **PROGRAMMING LANGUAGE:**

  Embedded C

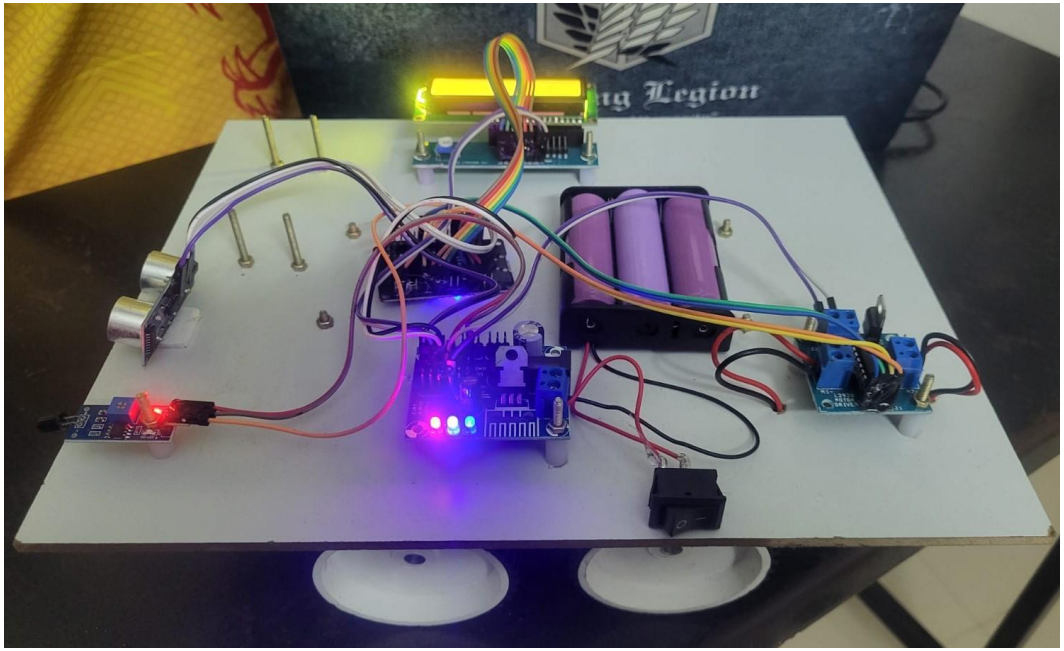
- **COMPILER :**

  Arduino IDE

- **MOBIL APP :**

  IoT module

OUTPUT



The system operates effectively by integrating sensors, a camera, and IoT communication to perform real-time surveillance. The ESP32 Microcontroller controls all functions, enabling obstacle detection, fire identification, and smooth robot movement. Live video streaming and remote monitoring are achieved through IoT connectivity, while the LCD displays system status. Overall, the robot performs reliably in hazardous environments and reduces the need for human intervention.

ADVANTAGES

- **Reduces Risk to Human Personnel**

  Operates in dangerous border and hazardous zones without direct human presence.

- **Real-Time Remote Monitoring**

  Enables live surveillance through IoT connectivity and mobile interface.

- **Live Video Streaming**

  Camera provides continuous visual feedback for better decision-making.

- **Multidirectional Mobility**

  Moves forward, backward, left, and right using DC motors and L293D driver.

- **Obstacle Detection & Collision Avoidance**

  Ultrasonic sensor prevents accidents during movement.

ADVANTAGES

- **Fire Detection Capability**

  Instantly detects flames or abnormal temperature conditions.

- **Human Presence Detection**

  Identifies intruders or trapped individuals in restricted areas.

- **Instant Alert Generation**

  Sends immediate notifications via cloud platform.

- **Cloud-Based Data Access**

  Allows monitoring and control from anywhere using internet.

- **Cost-Effective Implementation**

  Uses widely available embedded components.

APPLICATIONS

- Rising security concerns and hazardous working conditions have increased the demand for automated surveillance systems.
- Conventional border monitoring methods rely heavily on human patrols, exposing personnel to threats such as intrusions, fire hazards, and hostile environments.
- Recent advancements in robotics, embedded systems, and IoT have enabled intelligent surveillance platforms capable of operating remotely.
- Robotic vehicles integrated with sensors and cameras can perform continuous monitoring without fatigue.
- IoT connectivity further allows real time data transmission and remote control. This project introduces an embedded robotic vehicle that combines mobility, sensing, and cloud based monitoring to enhance border security and hazardous zone surveillance while reducing human risk.

CONCLUSION

- Successfully integrates embedded systems, IoT, and sensor technology for smart surveillance.
- Enables real-time monitoring with fire detection, obstacle sensing, and live video streaming.
- Provides remote-controlled mobility using ESP32 and L293D motor driver.
- Reduces human risk in hazardous and high-security environments.
- Cost-effective, scalable, and suitable for modern security applications.

REFERENCES

- Danish S; Samee Ullah Khan; Muhammad Attique Khan (2026), “Vision-based Fire Management System Using Autonomous Unmanned Aerial Vehicles: A Comprehensive Survey”, Artif. Intell. Rev.
- Akilan T; Satyam Chaudhary; Princi Kumari; Utkarsh Pandey (2025), “Surveillance Robot in Hazardous Place Using IoT Technology”, IEEE Conference Proceedings.
- Manesh T; Akshaya Jayaraj; Alen Joby; Abhilash Asokan (2025), “AI-Enabled Robotic Surveillance System for Critical Infrastructure”, Proc. ETIS 2025, IEEE.
- Gawan J S; Abhishek Meshram; Prajakta Shende; Shweta Pathe (2024), “IoT Based Wireless Surveillance Robot: Enhanced Control and Real-Time Video Streaming for Espionage and Security Applications”, Int. J. for Research in Applied Science & Engg. Tech., Vol.12, No.3.
- Meera M M R; Partho Sharothi Chowhan; Farah Alam Mim (2024), “Artificially Intelligent Surveillance and Security Sentinel for Technologically Enhanced and Protected Communities”, Int. J. Eng. & Manufacturing, Vol.14, No.4.

THANK YOU